ECONOMETRICS I
Practice Problem Set 3
(chapters 4-5-6)

1. Consider the following regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

We want to test $H_0: \ \beta_1 - 2\beta_2 = 1$.

(a) Let $\hat{\beta}_1$ and $\hat{\beta}_2$ be OLS estimators for β_1 and β_1. Find $Var(\hat{\beta}_1 - 2\hat{\beta}_2)$ in terms of the variances of $\hat{\beta}_1$ and $\hat{\beta}_2$ and the covariance between them.

(b) Write the t statistics for $H_0: \ \beta_1 - 2\beta_2 = 1$.

(c) Define $\theta_1 = \beta_1 - 2\beta_2$ and its OLS estimator $\hat{\theta}_1 = \hat{\beta}_1 - 2\hat{\beta}_2$. Write a regression equation that allows you to obtain $\hat{\theta}_1$ and its standard error.

2. Using the data in RDCHEM, the following equation was obtained by OLS:

$$\widehat{rdintens} = \underset{(.429)}{2.613} + \underset{(.00014)}{.00030}sales - \underset{(.0000000037)}{.0000000070}sales^2$$
$$n = 32, R^2 = .1484.$$

(a) At what point does the marginal effect of *sales* on *rdintens* become negative?

(b) Would you keep the quadratic term in the model? Explain.

(c) Define *salesbil* as sales measured in billions of dollars: $salesbil = sales/1,000$. Rewrite the estimated equation with *salesbil* and $salesbil^2$ as the independent variables. Be sure to report standard errors and the R-squared. [Hint: Note that $salesbil^2 = sales^2/(1,000)^2$.]

(d) For the purpose of reporting the results, which equation do you prefer?

3. The following model allows the return to education to depend upon the total amount of both parents' education, called *pareduc*:

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 educ \cdot pareduc + \beta_3 exper + \beta_4 tenure + u.$$

(a) Show that, in decimal form, the return to another year of education in this model is

$$\Delta \log(wage)/\Delta educ = \beta_1 + \beta_2 pareduc.$$

What sign do you expect for β_2? Why?

(b) Using the data in WAGE2, the estimated equation is

$$\widehat{\log(wage)} = \underset{(.13)}{5.65} + \underset{(.010)}{.047}\,educ + \underset{(.00021)}{.00078}\,educ.pareduc +$$
$$\underset{(.004)}{.019}\,exper + \underset{(.003)}{.010}\,tenure$$
$$n = 722, R^2 = .169.$$

(Only 722 observations contain full information on parents' education.) Interpret the coefficient on the interaction term. It might help to choose two specific values for *pareduc*-for example, $pareduc = 32$ if both parents have a college education, or $pareduc = 24$ if both parents have a high school education-and to compare the estimated return to *educ*.

(c) When *pareduc* is added as a separate variable to the equation, we get:

$$\widehat{\log(wage)} = \underset{(.38)}{4.94} + \underset{(.027)}{.097}\,educ + \underset{(.017)}{.033}\,pareduc - \underset{(.0012)}{.0016}\,educ.pareduc$$
$$+ \underset{(.004)}{.020}\,exper + \underset{(.003)}{.010}\,tenure$$
$$n = 722, R^2 = .174.$$

Does the estimated return to education now depend positively on parent education? Test the null hypothesis that the return to education does not depend on parent education.

4. Consider the following regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 + u$$

We want to test $H_0 : \ \beta_1 = 1, \ \beta_2 = -1, \ \beta_3 = 0$.

(a) Can you test this null hypothesis by t test? explain.

(b) Write the restricted model under the null hypothesis.

(c) Describe the F test procedure. Which version of the F statistic would you use, SSR version or the R^2 version?

(d) Describe the LM test procedure for this null hypothesis. Write the steps.

(e) Suppose that for a sample of size $n = 155$ we estimated the model above and obtained SSR=200. We also estimated the restricted model and obtained SSR = 220. Find the F statistic and state its distribution. What's your decision at 1% significance level? (critical value at $\alpha = 0.01$ is 3.78).

(f) Suppose that we only estimated the restricted model and obtained the residuals. Then we regressed these residuals on all x variables in the model and obtained the coefficient of determination, $R^2_{residuals} = 0.1$. Computed the LM test, state its distribution and your decision at 1% level. (critical value is 11.34 at $\alpha = 0.01$ significance level).

Review Questions:
(These will not be solved. Read the classnotes and the textbook)

5. If we want to use t and F tests in any sample size, then we need the normality assumption. True or False? Briefly explain.

6. OLS t statistics does not have the usual t distribution under the null hypothesis when there is heteroskedasticity. True or False? Explain.

7. OLS t statistics does not have the usual t distribution under the null hypothesis if an important variable is omitted. True or False? Explain.

8. Briefly explain the consistency of the OLS estimator. List and briefly explain the MLR assumptions required for the consistency.

9. Summarize the considerations when using logarithms in regression analysis (hint: read the text p. 194-5).

10. (COMPUTER EXERCISE - Use R to solve) Use the housing price data in HPRICE1 for this exercise.

(a) Estimate the model

$$\log(price) = \beta_0 + \beta_1 \log(lotsize) + \beta_2 \log(sqrft) + \beta_3 bdrms + u$$

and report the results in the usual OLS format.

(b) Find the predicted value of $\log(price)$, when $lotsize = 20,000$, $sqrft = 2,500$, and $bdrms = 4$. Using the methods in Section 6-4, find the predicted value of *price* at the same values of the explanatory variables.

(c) For explaining variation in *price*, decide whether you prefer the model from part (i) or the model

$$price = \beta_0 + \beta_1 lotsize + \beta_2 sqrft + \beta_3 bdrms + u.$$